

PageRank Equations for Each Page

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$$PR(p) = \frac{1-d}{N} + d \sum_{q \in M(p)} \frac{PR(q)}{L(q)}$$

◆ Page A:

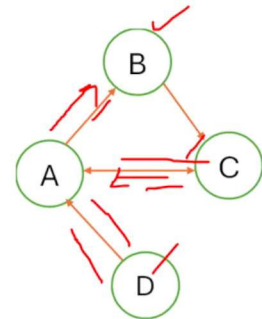
Incoming links: C → A, D → A $PR(A) = 0.0375 + 0.85 \left(\frac{PR(C)}{1} + \frac{PR(D)}{1} \right)$

◆ Page B:

Incoming link: A → B $PR(B) = 0.0375 + 0.85 \left(\frac{PR(A)}{2} \right)$

◆ Page C:

Incoming links: A → C, B → C $PR(C) = 0.0375 + 0.85 \left(\frac{PR(A)}{2} + \frac{PR(B)}{1} \right)$



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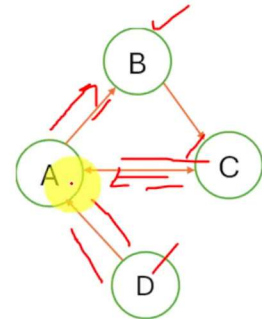
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◆ Page C:

Incoming links: A → C, B → C $PR(C) = 0.0375 + 0.85 \left(\frac{PR(A)}{2} + \frac{PR(B)}{1} \right)$



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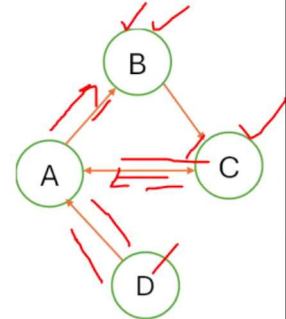
Incoming link: A → B $PR(B) = 0.0375 + 0.85 \left(\frac{PR(A)}{2} \right)$

◆ Page C:

Incoming links: A → C, B → C $PR(C) = 0.0375 + 0.85 \left(\frac{PR(A)}{2} + \frac{PR(B)}{1} \right)$

◆ Page D:

Incoming links: None $PR(D) = 0.0375 + 0.85 \cdot 0 = 0.0375$



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$$PR(p) = \frac{1-d}{N} + d \sum_{q \in M(p)} \frac{PR(q)}{L(q)}$$

◆ Page A:

Incoming links: C → A, D → A $PR(A) = 0.0375 + 0.85 \left(\frac{PR(C)}{1} + \frac{PR(D)}{1} \right)$

◆ Page B:

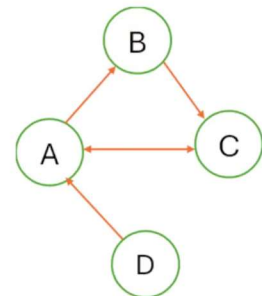
Incoming link: A → B $PR(B) = 0.0375 + 0.85 \left(\frac{PR(A)}{2} \right)$

◆ Page C:

Incoming links: A → C, B → C $PR(C) = 0.0375 + 0.85 \left(\frac{PR(A)}{2} + \frac{PR(B)}{1} \right)$

◆ Page D:

Incoming links: None $PR(D) = 0.0375 + 0.85 \cdot 0 = 0.0375$



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PageRank Algorithm: Iteration 1 (Using Initial Values)

$$PR(A) = 0.0375 + 0.85 \left(\frac{0.25}{1} + \frac{0.25}{1} \right)$$

$$= 0.0375 + 0.85(0.5) = 0.0375 + 0.425 = 0.4625$$

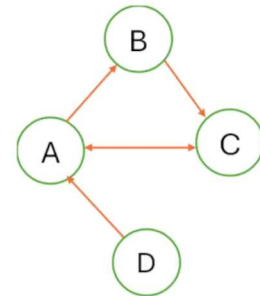
$$PR(B) = 0.0375 + 0.85 \left(\frac{0.25}{2} \right)$$

$$= 0.0375 + 0.85(0.125) = 0.0375 + 0.10625 = 0.14375$$

$$PR(C) = 0.0375 + 0.85 \left(\frac{0.25}{2} + \frac{0.25}{1} \right)$$

$$= 0.0375 + 0.85(0.125 + 0.25) = 0.0375 + 0.31875 = 0.35625$$

$$PR(D) = 0.0375 + 0.85(0) = 0.0375$$

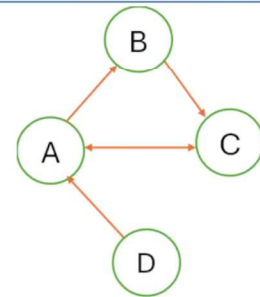


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PageRank Algorithm: Iteration 4 to until Convergence

Iteration	PR(A)	PR(B)	PR(C)	PR(D)
0	0.25	0.25	0.25	0.25
1	0.4625	0.14375	0.35625	0.0375
2	0.37219	0.23406	0.35625	0.0375
3	0.37219	0.19568	0.39463	0.0375
4	0.37161	0.19566	0.39522	0.0375
5	0.37161	0.19557	0.39532	0.0375
6	<u>0.37162</u>	<u>0.19557</u>	<u>0.39531</u>	<u>0.0375</u>



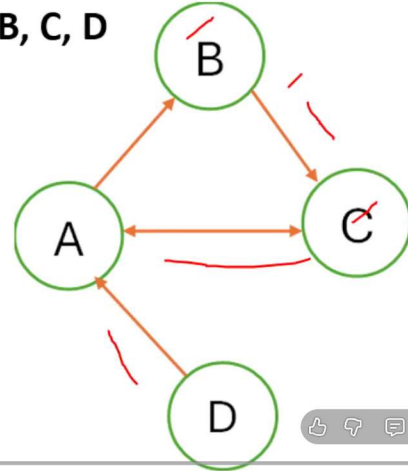
- Convergence** is reached when the values stop changing significantly (≤ 0.0001 difference between iterations).

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Page Rank Algorithm – Solved Example – 1

- Here we will apply page rank algorithm and find the final page rank values for the following given pages → **A, B, C, D**
- $A \rightarrow B, C$
- $B \rightarrow C$
- $C \rightarrow A$
- $D \rightarrow A$
- Consider Damping factor: $d = 0.85$



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Page Rank Algorithm

- PageRank is an algorithm originally developed by Larry Page and Sergey Brin (Google's founders) to rank web pages in search engine results.
- **Core idea:** A page is important if it is linked to by other important pages.

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